Ordered butterflies and oval constructions for $\mathrm{PSL}_3(4)$

L 100 finite construction certificate

We certify seven concrete constructions in the fixed 22-page Halle preprint of Gottschalk and Leemans [1], using the already certified point action of $G = \mathrm{PSL}_3(4)$ and its coset geometries. The reference is the preprint, not an independently collated journal edition. We make the butterfly convention and the otherwise unspecified oval classes explicit. This is a construction component, not a new classification or an assertion that all remaining structural names in the source have been proved.

Acquired finite premises. The coordinate model is the plane over $\mathbb{F}_2[t]/(t^2 + t + 1)$, with an explicit bijection from its normalized coordinate vectors to the retained 21-point permutation model. Its saved 20160 permutations and two generators are the same literal group as the coset catalogue. The acquired point/line/hyperoval certificate gives the 21 lines and three 56-element hyperoval orbits. The acquired Baer certificate gives the three 120-element orbits $\mathcal{B}_0, \mathcal{B}_1, \mathcal{B}_2$ of Fano subplanes. Completeness of these 360 subplanes follows from the frame count $21 \cdot 20 \cdot 16 \cdot 9/(7 \cdot 6 \cdot 4) = 360$, with a unique embedded Fano plane through a frame. These literal inputs, their proofs and their distinct reviews are acquired without repeating a group or subgroup census.

An explicit convention. An ordered butterfly in this note is an ordered pair of unordered poles $b = (\{B_1, B_2\}, \{C_1, C_2\})$, consisting of four distinct subplanes, such that, for a point p and a line z,

$$\begin{aligned} B_i \cap C_j = \{p\} \quad (1 \le i, j \le 2), \qquad T_+ = B_1 \cap B_2, \quad T_- = C_1 \cap C_2, \\ |T_+| = |T_-| = 3, \qquad T_+ \cup T_- = z. \end{aligned} \tag{1}$$

This all-cross-intersections convention is stronger than the two displayed cross-intersection conditions on source page 4; the distinction is addressed on the last page. Write $U_+ = B_1 \cup B_2$, $U_- = C_1 \cup C_2$ and

$$Q_+ = T_+ \setminus \{p\}, \qquad L_\pm = \{l : p \in l \subseteq U_\pm\}, \qquad H_+ = \{(B_i \setminus z) \cup (T_- \setminus \{p\}) : i = 1, 2\},$$

with H_- defined by interchanging the poles. Thus a line "contained in a pole" means containment in the union of its two point sets. The central line is not such a line: its intersection with that union has only three points.

Lemma 1. *For each $\mathcal{B}_i$, (1) gives exactly 1260 ordered butterflies, one orbit of G, with stabilizer $D_8 \times C_2$. Both L_+ and L_- have two lines. The four members of $H_+ \cup H_-$ are distinct hyperovals in one orbit $\mathcal{H}_{e(i)}$, and $\mathcal{B}_i$ meets every hyperoval in this orbit evenly. The positive pole determines the ordered butterfly uniquely. Its unordered four-subplane set is also determined by an opposite pair of subplanes or by one of its subplanes together with (p, z).*

Proof. The finite certificate has a complete, small domain. Test all $\binom{360}{2} = 64620$ subplane pairs. Retain a pair when its intersection is a collinear triple T. There are 3780 such pairs. For each, its line z is unique. For each of the three choices $p \in T$, test all retained pairs whose intersection is $(z \setminus T) \cup \{p\}$, and impose all four equalities in (1). This exhausts every possible opposite pole. The combined search over these three choices gives exactly one opposite pole for each retained pair. The resulting objects have 1260 members in each Baer orbit. Closure and transitivity are

checked using the two actual group generators. The alternative uniqueness assertions are checked by grouping the full list by the indicated exact subplane/flag keys, never by numerical invariants.

For a representative in each orbit, test all saved group permutations for stabilization of the ordered poles. The 16 retained permutations are verified closed. The certificate exhibits a, b, z with $|a| = 4$, $|b| = |z| = 2$, $bab = a^{-1}$ and $[a, z] = [b, z] = 1$; its sixteen words $a^i b^j z^k$, $0 \le i < 4$, $0 \le j, k < 2$, are distinct and exhaust the stabilizer. This proves its asserted structure, not merely its order. The source's auxiliary subplane-stabilizer transitivity is checked on all 21 partners of one subplane in each orbit.

Each asserted hyperoval is checked against the complete acquired list. All $168\binom{6}{4} = 2520$ four-subsets of these hyperovals are distinct; therefore the associated quadrangle has a unique such completion. Containment tests give the two pole-line sets, the even-intersection assertion and all stated uniqueness keys. These are finite set tests on the complete domains just described; the replay recomputes them. □

The six butterfly geometries. For $B \in \mathcal{B}_i$, set $R(B, b) \iff T_+ \subseteq B$ and $B \notin \{B_1, B_2\}$. The following table states every incidence relation between distinct types. Here h, h' range over two tagged copies of the same class $\mathcal{H}_{e(i)}$, x is a point, l a line, and b an ordered butterfly from $\mathcal{B}_i$. The ordered types and the six entries are respectively the relations 12,13,14,23,24,34. Equal-type incidence is equality.

Source	Types	Relations in the stated order
4.2	l, x, h, b	$x \notin l;\ l \cap h = \varnothing;\ l \in L_+;\ x \in h;\ x \in Q_+;\ h \in H_-$
4.3	h, h', B, b	$h \cap h' = \varnothing;\ \lvert h \cap B\rvert = 4;\ h \in H_-;$
		$h' \cap B = \varnothing;\ h' \in H_+;\ R(B, b)$
4.4	h, h', x, b	$h \cap h' = \varnothing;\ x \in h;\ h \in H_-;\ x \notin h';\ h' \in H_+;\ x \in Q_+$
4.5	l, x, h, b	$x \notin l;\ l \cap h \neq \varnothing;\ l \in L_-;\ x \in h;\ x \in Q_+;\ h \in H_-$
4.6	x, h, B, b	$x \in h;\ x \in B;\ x \in Q_+;\ \lvert h \cap B\rvert = 4;\ h \in H_-;\ R(B, b)$
4.7	l, x, B, b	$x \notin l;\ \lvert l \cap B\rvert = 3;\ l \in L_+;\ x \in B;\ x \in Q_+;\ B \in \{B_1, B_2\}$

These are the native signs on pages 16–18. In particular 4.5 uses type-one hyperovals from 4.4 and the negative pole for its line relation.

Oval and dual-oval classes. Use the coordinate polarity $[a] \mapsto \{[v] : a \cdot v = 0\}$ with its literal transport to the saved point and line labels. It induces a permutation d of the three Baer classes. For each hyperoval orbit, delete each of the six points from each hyperoval. This gives 336 distinct ovals in one orbit; the deleted point is its nucleus, since the five tangent lines are checked to pass through it. Polarity gives the corresponding dual ovals, with a nucleus line. For each ordered pair $i \neq j$, choose

$$B \in \mathcal{B}_i, \quad C \in \mathcal{B}_j, \quad O \in \mathrm{Ovals}(\mathcal{H}_{e(i)}), \quad D \in \mathrm{DualOvals}(\mathcal{H}_{e(d(j))}). \tag{2}$$

In the saved class ordering, $e = (0, 1, 2)$ and $d = (0, 2, 1)$. The indices are part of the explicit model, not intrinsic names. Writing $n(O)$ for a point and $n(D)$ for a line, the source 4.11 relations are

$$\begin{aligned}
B \sim C &\iff B \cap C = \varnothing,\\
B \sim O &\iff |B \cap O| = 4,\\
B \sim D &\iff |B \cap n(D)| = 3 \text{ and } |B \cap l| = 1 \ (l \in D),\\
C \sim O &\iff n(O) \in C \text{ and } C \cap O = \varnothing,\\
C \sim D &\iff \#\{l \in D : |C \cap l| = 3\} = 4,\\
O \sim D &\iff |O \cap n(D)| = 2 \text{ and } (O \setminus n(D)) \cap l = \varnothing \ (l \in D).
\end{aligned}$$

All six choices in (2) are certified. The source names an oval class and a dual-oval class without specifying their relative identification; (2) supplies a successful precise choice, not a claim about arbitrary choices.

Theorem 2. *For each of the three Baer classes, the six butterfly recipes above are equivariantly isomorphic, up to an explicitly given automorphism of G and type permutation, to source geometries 4.2–4.7. All six choices in (2) similarly give geometry 4.11. Consequently these constructions are firm, residually connected, satisfy $(IP)_2$, and have a flag-transitive RWPRI G-action, with the local action qualifications stated below.*

Proof. We use the accepted coset catalogue, not a new recognition by orders or diagrams. The source-to-catalogue indices are respectively $24, 19, 22, 23, 18, 17, 25$. For each of the 24 maps, the certificate records a type permutation, one base object fixed by each actual parabolic, an explicit group automorphism α, and every object image. The map on right cosets is $P_t g \mapsto X_t^{\alpha(g)}$. Every one of the 20160 group entries in each of the four coset partitions is tested: all members of a coset give the recorded object. The recorded maps are bijections on each type. For every cross-type pair, incidence and nonincidence agree exactly; the two generator actions agree on every object. Thus these are actual typed incidence and action isomorphisms.

The automorphisms used are explicit point normalizers and coordinate duality. Point normalizers are checked on the whole saved group; duality is checked on every generator edge of its Cayley graph. The actual coordinate-to-retained label bijection is always used. None of these maps is inferred from a name, group order, or an unlabelled graph comparison.

The cited geometric properties, including extension of every partial flag, are acquired for these exact coset geometries. Isomorphisms transport all flags, chambers, residues and stabilizer actions, proving the assertions for the concrete constructions. Local double transitivity is not used as a recognition filter: in the displayed source type order,4.2 has flags $(1, 1, 1, 0)$,4.11 has $(1, 1, 0, 0)$, and 4.3–4.7 have four true flags. These follow from the acquired literal panel actions and the recorded type maps. □

A narrow qualification to the printed butterfly description. On page 4 the source displays only $B_1 \cap B_1' = B_2 \cap B_2' = \{p\}$, together with the triple/line condition. Those two displayed requirements alone do not imply the next sentence's uniqueness from (B_1, B_2). Here is a direct witness in the retained point labels. Fix

$$B_1 = \{1, 2, 3, 6, 7, 10, 11\}, \qquad B_2 = \{1, 4, 5, 8, 9, 10, 11\}.$$

Two different completions, in the same Baer orbit as B_1, B_2, are

p	B_1'	B_2'
1	$\{1, 12, 13, 15, 16, 18, 20\}$	$\{1, 12, 13, 14, 17, 19, 21\}$
10	$\{4, 8, 10, 12, 13, 16, 19\}$	$\{3, 7, 10, 12, 13, 14, 18\}$.

Both have central line $\{1, 10, 11, 12, 13\}$ and satisfy exactly the two printed cross equalities. In the first, all four cross intersections have size 1; in the second their sizes, by rows, are $(1, 3)$ and $(3, 1)$. All six point sets are checked against the complete saved subplane list; actual permutations of G mapping B_1 to each one are supplied. Hence neither same-orbit restrictions nor forgetting the pole ordering remove the ambiguity. This establishes insufficiency of the displayed conditions, not a claim that the conventional butterflies in the source's cited literature have that broader meaning. We do not inspect or contradict that external convention here. Convention(1) makes the intended constructions and the stabilizer assertion completely explicit. No principal classification is refuted by this ancillary qualification.

Reproducibility and attribution. The packet retains the initial failed uniqueness assertion under the weaker displayed rule, its diagnosis, and the successful successor. Default replay checks finite object completeness and the 24 supplied maps; it does not search for maps, rerun a subgroup census or recompute graph automorphism groups. The names and recipes are due to Gottschalk–Leemans and their cited antecedents. This certificate gives literal construction joins and an explicit scope qualification, with no novelty, priority, measurement or whole-source claim.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL$(3, 4)$, Halle preprint 99–30, fixed 22-page source, pp. 4–5,16–19.